Software Requirements Specification (SRS)

Project: Online Bookstore
Technology: Python (Django)
Version: 1.0
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Revision history

Version	Date	Author	Change summary	Approval
1.0	06-08-2025	Vihaan Wadhwa	SRS with diagrams embedded	

Approvals

Role	Name	Signature / Email	Date
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Table of Contents

1. Introduction
2. Overall description
3. External interfaces
4. System features (detailed)
5. Non-functional requirements (detailed)
6. Quality attributes & Acceptance tests
7. UML Use-Case Diagram
8. Requirements Traceability Matrix (RTM)

1. Introduction

1.1 Purpose

This document is a Software Requirements Specification (SRS) for the Online Bookstore, a web-based e-commerce platform for buying and selling books. It defines the functional and non-functional requirements, external interfaces, and acceptance/verification criteria intended to guide design, development, and testing of the system.

1.2 Scope

Covers buyer, seller, and administrator interactions on the platform: account registration and authentication, browsing and searching the book catalogue, cart and checkout, order placement and tracking, seller book-listing management, administrative inventory/user management, and reviews/ratings. Excludes the internal implementation of third-party services (payment gateway, email/SMS provider, shipping/logistics partner) beyond their published APIs, and excludes physical warehouse or courier operations.

1.3 Audience

Developers, QA/Testers, Project Evaluators, Course Instructor, and future maintainers of the system.

1.4 Definitions

List of acronyms: SRS (Software Requirements Specification), UI (User Interface), API (Application Programming Interface), ORM (Object-Relational Mapping), JWT (JSON Web Token), HTTPS/TLS (Hypertext Transfer Protocol Secure / Transport Layer Security), RBAC (Role-Based Access Control), PCI-DSS (Payment Card Industry Data Security Standard), WCAG (Web Content Accessibility Guidelines), RTO/RPO (Recovery Time/Point Objective).

2. Overall description

2.1 Product perspective

The Online Bookstore is a self-contained, web-based client-server application. It consists of a browser-based front end (server-rendered templates using Django), a back-end application server exposing a REST API, a relational or document database for persistent storage, and integrations with an external payment gateway and email/notification service.

2.2 Major product functions (detailed)

- User registration, login, and profile management
- Browse and search the book catalogue by title, author, ISBN, category, and price
- View detailed book information, including price, stock, description, and seller
- Shopping cart management (add, update quantity, remove)
- Checkout with address selection and payment processing
- Order placement, order history, and shipment tracking
- Seller book-listing management (create, update, remove listings)
- Administrative management of users, listings, and inventory
- Product reviews and ratings
- Wishlist management and email notifications for order/account events

2.3 User roles and characteristics

- Buyer: registered customer who browses, purchases, and tracks orders; assumed to have basic web/e-commerce familiarity.
- Seller: registered vendor who lists and manages their own books for sale; needs simple listing and order-fulfilment tools.
- Admin: platform operator responsible for approving listings, managing users/inventory, and monitoring platform health.

2.4 Operating environment

Back end built with Python/Django with PostgreSQL. Front end is a responsive web application accessible from modern desktop and mobile browsers; deployment on a cloud platform (e.g., Render, Heroku, AWS) with HTTPS enabled.

2.5 Constraints

- Academic mini-project timeline and team size limit scope to the features listed above.
- Must be implemented using Python/Django.
- Payment processing uses a sandbox/test-mode payment gateway integration (no real transactions).
- Must follow basic data-privacy practices for user and payment-related data.

3. External interface requirements

3.1 User interfaces

Primary UI: responsive web interface usable on desktop and mobile browsers, with clear navigation between catalogue, cart, checkout, and account/order pages. Accessibility: readable contrast, keyboard-navigable forms, and descriptive labels for screen readers.

3.2 Hardware interfaces

No specialised hardware is required. The system is accessed from standard client devices (desktop, laptop, tablet, or smartphone) equipped with a modern web browser and internet connectivity.

3.3 Software interfaces

- Payment Gateway API (e.g., Razorpay/Stripe sandbox) over HTTPS for processing and confirming payments.
- Email/Notification service API (e.g., SMTP or a transactional email provider) for order confirmations and account alerts.
- Database interface: PostgreSQL via Django ORM.
- Optional shipping/logistics API for shipment status updates.

3.4 Communications

- HTTPS/TLS 1.2+ enforced for all client-server and server-to-third-party communication.
- REST API using JSON payloads between front end and back end.
- Retry/error-handling strategy for failed third-party (payment/notification) calls.

4. System features (detailed)

Each requirement below includes its type, priority, source, and acceptance criteria / test case reference. IDs follow BOOK-F-###.

4.1 Authentication & Account Management

Description: Allow buyers and sellers to register, log in securely, and manage their account/password.

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
BOOK-F-001	The system shall allow a new user to register with a unique email address, password, and role (Buyer/Seller).	Functional	High	Buyer / Seller	AC-BOOK-F-001: Valid unique email + password creates an active account. Test: TC-Auth-01	Email uniqueness check
BOOK-F-002	The system shall authenticate users via email and password, and issue a session/JWT token on success.	Functional	High	Buyer / Seller / Admin	AC-BOOK-F-002: Correct credentials return a valid token; incorrect credentials are rejected. Test: TC-Auth-02	Depends on BOOK-F-001
BOOK-F-003	The system shall lock a user account for 15 minutes after 5 consecutive failed login attempts.	Functional	Medium	Security	AC-BOOK-F-003: 6th consecutive failed attempt returns a lockout message. Test: TC-Auth-03	Related to BOOK-SR-005
BOOK-F-004	The system shall allow a user to reset a forgotten password via a time-limited email link.	Functional	Medium	Buyer / Seller	AC-BOOK-F-004: Reset link expires after 30 minutes and allows setting a new password once. Test: TC-Auth-04	Requires email service

4.2 Book Catalogue & Search

Description: Allow buyers to browse and search the book catalogue and view detailed listings.

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
BOOK-F-005	The system shall display a paginated catalogue of books with title, cover image, price, and availability.	Functional	High	Buyer	AC-BOOK-F-005: Catalogue page loads with correct pagination and data. Test: TC-Cat-01	
BOOK-F-006	The system shall allow searching books by title, author, ISBN, or keyword, with filters for category and price range.	Functional	High	Buyer	AC-BOOK-F-006: Search returns matching results within 2 seconds for typical queries. Test: TC-Cat-02	
BOOK-F-007	The system shall display a detailed book page showing description,	Functional	High	Buyer	AC-BOOK-F-007: Book detail page shows all required fields	Depends on BOOK-F-016

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
	price, stock, seller, and average rating.				accurately. Test: TC-Cat-03	

4.3 Shopping Cart & Checkout

Description: Allow buyers to manage a cart and complete purchases securely.

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
BOOK-F-008	The system shall allow a buyer to add, update the quantity of, or remove books from a shopping cart.	Functional	High	Buyer	AC-BOOK-F-008: Cart total updates correctly after each action. Test: TC-Cart-01	
BOOK-F-009	The system shall allow a buyer to select/enter a shipping address and a payment method at checkout.	Functional	High	Buyer	AC-BOOK-F-009: Checkout requires a valid address and payment method before proceeding. Test: TC-Cart-02	
BOOK-F-010	The system shall process payment through the payment gateway and generate an order confirmation only on success.	Functional	High	Buyer / Business	AC-BOOK-F-010: Successful payment creates an order and confirmation email; failed payment leaves cart unchanged. Test: TC-Cart-03	Requires Payment Gateway API

4.4 Order Management & Tracking

Description: Allow buyers to view and manage their orders after purchase.

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
BOOK-F-011	The system shall allow a buyer to view their order history with status (placed, shipped, delivered, cancelled).	Functional	High	Buyer	AC-BOOK-F-011: Order list reflects current status accurately. Test: TC-Ord-01	
BOOK-F-012	The system shall update and display shipment tracking status for each order.	Functional	Medium	Buyer / Operations	AC-BOOK-F-012: Status changes (e.g., shipped -> delivered) are reflected within one refresh. Test: TC-Ord-02	Optional shipping API
BOOK-F-013	The system shall allow a buyer to cancel an order or request a return within a configurable policy window (e.g., 7 days).	Functional	Medium	Buyer	AC-BOOK-F-013: Cancellation/return request is rejected after the policy window elapses. Test: TC-Ord-03	

4.5 Seller & Admin (Inventory) Management

Description: Allow sellers to manage their listings and admins to oversee the platform.

Req ID	Requirement (shall...)	Type	Priority	Source/Stake holder	Acceptance criteria / Test case ref	Comments / Dependencies
BOOK-F-014	The system shall allow a seller to create a new book listing with title, price, stock quantity, and images.	Functional	High	Seller	AC-BOOK-F-014: New listing appears in catalogue after admin approval. Test: TC-Sell-01	Depends on BOOK-F-016 (admin approval)
BOOK-F-015	The system shall allow a seller to update or remove their own book listings.	Functional	High	Seller	AC-BOOK-F-015: Only the owning seller (or admin) can edit/delete a given listing. Test: TC-Sell-02	Related to BOOK-SR-004 (RBAC)
BOOK-F-016	The system shall allow an admin to approve or reject new/updated seller listings before they go live.	Functional	High	Admin	AC-BOOK-F-016: Rejected listings are not visible in the public catalogue and the seller is notified. Test: TC-Admin-01	
BOOK-F-017	The system shall allow an admin to view and adjust overall inventory levels and deactivate problematic listings or accounts.	Functional	Medium	Admin	AC-BOOK-F-017: Deactivated listing/account is immediately inaccessible to buyers. Test: TC-Admin-02	

4.6 Reviews, Ratings, Wishlist & Notifications

Description: Support post-purchase engagement features.

Req ID	Requirement (shall...)	Type	Priority	Source/Stake holder	Acceptance criteria / Test case ref	Comments / Dependencies
BOOK-F-018	The system shall allow a buyer who has purchased a book to submit a rating (1-5) and written review.	Functional	Medium	Buyer	AC-BOOK-F-018: Review is stored and reflected in the book's average rating. Test: TC-Rev-01	Depends on BOOK-F-011 (purchase history)
BOOK-F-019	The system shall allow a buyer to add or remove books from a personal wishlist.	Functional	Low	Buyer	AC-BOOK-F-019: Wishlist persists across sessions for a logged-in user. Test: TC-Rev-02	
BOOK-F-020	The system shall send an email notification to the buyer on order confirmation, shipment, and delivery status changes.	Functional	Medium	Buyer	AC-BOOK-F-020: Notification email is sent within 1 minute of each status change. Test: TC-Rev-03	Requires email service

5. Non-functional requirements (detailed)

NFRs below are measurable and tied to test plans. IDs follow BOOK-NF-###.

Req ID	Requirement	Category	Priority	Acceptance criteria / Measurement
BOOK-NF-001	Catalogue and search pages shall load within 3 seconds, and checkout shall complete within 5 seconds, for 90% of requests under normal load.	Performance	High	90th percentile response times meet target in load test. Test: TC-Perf-01
BOOK-NF-002	The system shall support at least 500 concurrent active users without service degradation.	Scalability	Medium	Load test with 500 concurrent simulated users shows no errors/timeouts. Test: TC-Perf-02
BOOK-NF-003	The platform shall provide 99.5% availability per month, excluding scheduled maintenance windows.	Reliability/Availability	High	Uptime monitoring reports show >=99.5% monthly. Test: Ops monitoring report
BOOK-NF-004	The web interface shall be responsive and usable on both desktop and mobile browsers, and shall conform to WCAG 2.1 AA where applicable.	Usability/Accessibility	Medium	Accessibility audit and responsive-layout checks pass. Test: TC-UX-01
BOOK-NF-005	Automated tests using Django's TestCase/pytest-django, covering at least 70% of core business logic.	Maintainability	Medium	Test coverage report shows >=70% coverage on core modules. Test: CI coverage report
BOOK-NF-006	Order and payment-related data shall be backed up daily with a Recovery Point Objective (RPO) of 24 hours and Recovery Time Objective (RTO) of 4 hours.	Reliability/Data Recovery	Medium	Backup restoration test meets RPO/RTO targets. Test: TC-Ops-02

5.1 Security

5.1.1 Security Objectives

This project defines the following security objectives:

- Protect the confidentiality of customer personal data and payment information, in line with PCI-DSS principles.
- Ensure the integrity of order and payment transactions so they cannot be tampered with in transit or storage.
- Ensure the availability of authentication and checkout functions against common abuse (e.g., credential stuffing, brute force).

5.1.2 Security Requirements

Req ID	Requirement (shall...)	Type	Priority	Acceptance criteria / Test case ref
BOOK-SR-001	TLS 1.2+ shall be mandatory for all client-server and server-to-third-party network connections.	Security	High	AC-BOOK-SR-001: Any plain-HTTP request is redirected/rejected. Test: TC-Sec-01

Req ID	Requirement (shall...)	Type	Priority	Acceptance criteria / Test case ref
BOOK-SR-002	Django's built-in password hashing (PBKDF2/Argon2 via django.contrib.auth) shall be used; passwords shall never be stored or logged in plaintext.	Security	High	AC-BOOK-SR-002: Database inspection shows only hashed values; logs contain no raw passwords. Test: TC-Sec-02
BOOK-SR-003	Payment card data shall not be stored on application servers; payment details shall be tokenised via the payment gateway (PCI-DSS SAQ-A aligned).	Security	High	AC-BOOK-SR-003: No raw card data present in application database or logs. Test: TC-Sec-03
BOOK-SR-004	The system shall enforce role-based access control (RBAC) so Buyer, Seller, and Admin actions are authorised server-side.	Security	High	AC-BOOK-SR-004: Attempting a Seller/Admin-only action as Buyer returns an authorisation error. Test: TC-Sec-04
BOOK-SR-005	Django's ORM parameterized queries and template auto-escaping shall be relied upon as the primary defense, supplemented by DRF serializer validation for API inputs.	Security	High	AC-BOOK-SR-005: Automated security scan shows no injection/XSS findings; login endpoint blocks after threshold. Test: TC-Sec-05

6. Quality attributes & Acceptance tests

- Exit criteria for acceptance: All high-priority functional requirements implemented and verified, no critical NFR or security-requirement failures, and the RTM shows all mapped test cases passed.

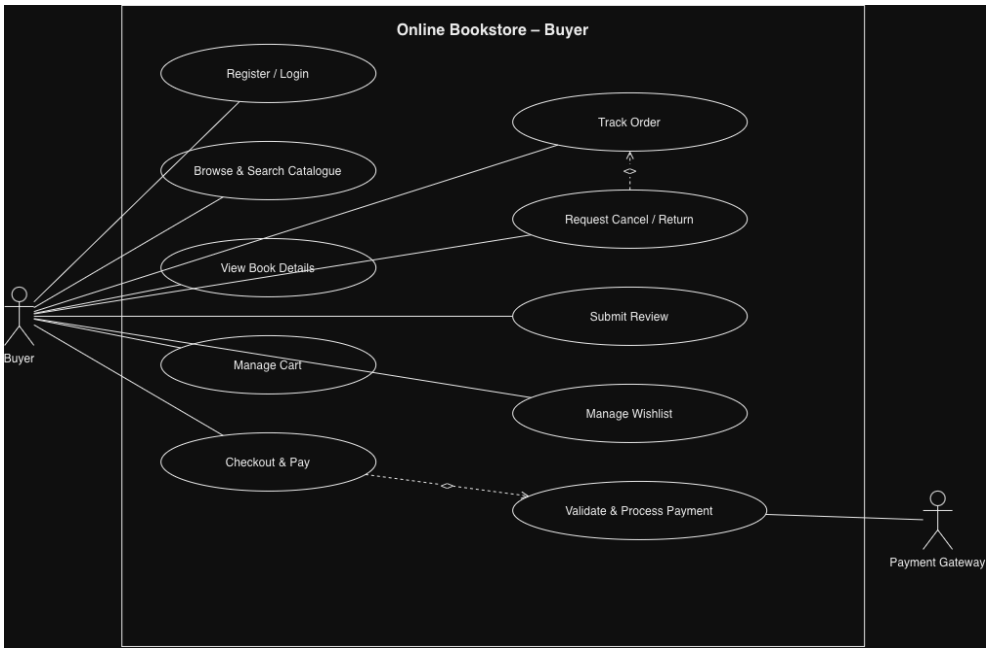
- Acceptance test suites: Authentication, Catalogue & Search, Cart & Checkout, Order Management, Seller & Admin Management, Reviews & Notifications, Performance, Security, and Accessibility tests.

7. System models and diagrams

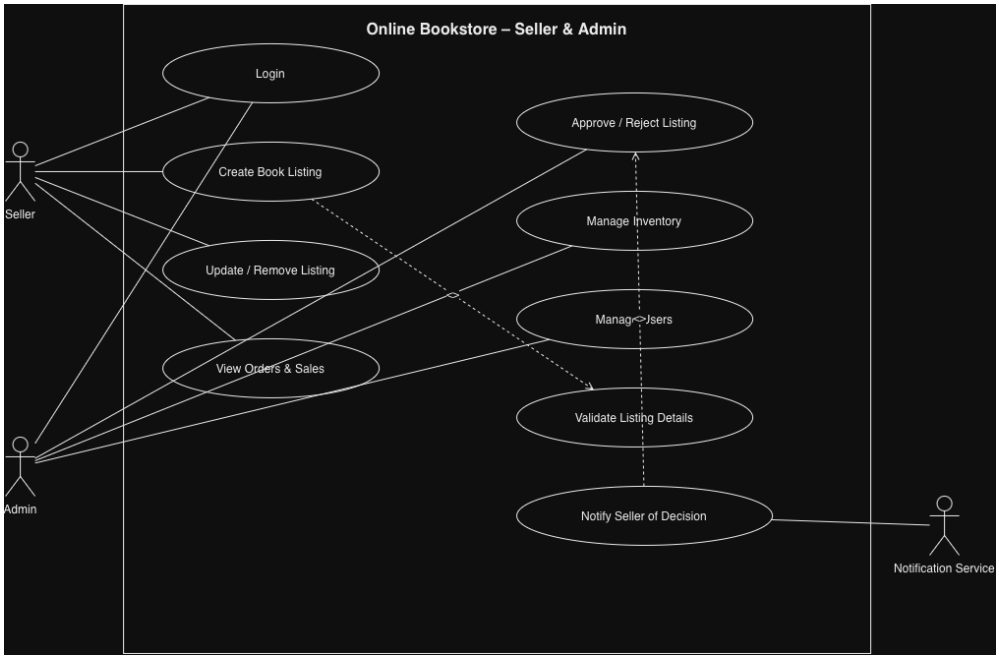
7.1 UML Use-Case diagram

At least 2 UML use-case diagrams shall be created for this project, for example:

- Buyer use-case diagram: Register/Login, Browse/Search Catalogue, Manage Cart, Checkout & Pay, Track Order, Submit Review, Manage Wishlist.



- Seller/Admin use-case diagram: Login, Create/Update Listing, Approve/Reject Listing (Admin), Manage Inventory (Admin), Manage Users (Admin).



8. Requirements Traceability Matrix (RTM)

Req ID	Requirement short	Section ref / Design Spec	Module	Test case(s)	Status (N/P/A)	Comments
BOOK-F-001	Register account	4.1 / DS-Auth-01	AuthModule	TC-Auth-01	N	
BOOK-F-002	Login & token issue	4.1 / DS-Auth-01	AuthModule	TC-Auth-02	N	
BOOK-F-008	Cart add/update/remove	4.3 / DS-Cart-01	CartModule	TC-Cart-01	N	
BOOK-F-010	Checkout & payment	4.3 / DS-Checkout-01	CheckoutModule / Payment API	TC-Cart-03	N	
BOOK-F-014	Seller create listing	4.5 / DS-Sell-01	SellerModule	TC-Sell-01	N	
BOOK-NF-001	Response time target	5 / DS-Perf-01	WebUI / API	TC-Perf-01	N	
BOOK-SR-002	Password hashing	5.1.2 / DS-Sec-01	AuthModule	TC-Sec-02	N	
BOOK-SR-004	RBAC enforcement	5.1.2 / DS-Sec-01	AuthModule / API Gateway	TC-Sec-04	N	